

DATA 255 Lab 1: Step 3 Analysis

Architecture Justification

- Designed for a small dataset (300 images/class, no pretrained weights)
- Channels double each block (32->64->128->256) while spatial size shrinks, balancing capacity against overfitting risk
- Global Average Pooling replaces flattening to avoid a ~57M-parameter FC layer that would overfit on 6,000 images, keeping the model at ~1.2M parameters
- Dropout (0.5, 0.3) regularizes the classifier head
- BatchNorm after every convolution stabilizes training

Final Architecture

Layer	Operations	Output
Input	-	240x240x3
Block 1	Conv(3->32) x2, BN, ReLU, MaxPool	120x120x32
Block 2	Conv(32->64) x2, BN, ReLU, MaxPool	60x60x64
Block 3	Conv(64->128) x2, BN, ReLU, MaxPool	30x30x128
Block 4	Conv(128->256) x1, BN, ReLU, MaxPool	15x15x256
Block 5	Conv(256->256) x1, BN, ReLU, GlobalAvgPool	1x1x256
Classifier	Dropout(0.5), Linear(256->128), ReLU, Dropout(0.3), Linear(128->20)	20

Total parameters: 1,209,652

Key Hyperparameters

- Optimizer: Adam (fixed lr = 0.001)
- Loss function: CrossEntropyLoss
- Batch size: 32
- Epochs: 44 (best checkpoint, used for ONNX/NPU deployment)

Preprocessing

- Train: Resize(260) -> RandomCrop(240) -> RandomHorizontalFlip -> ColorJitter -> Normalize (ImageNet mean/std)
- Val: Resize(240) -> Normalize (same mean/std, no augmentation)

Overall Accuracy

- PyTorch (validation set, checkpoint used for NPU deployment): 73.30%
- NPU (.mxq, quantized): 74.20% (742/1000)
- Note: the training log recorded 71.40% at the moment this checkpoint (epoch 44) was saved; re-measuring the same saved weights afterward gave 73.30%. This small run-to-run gap is attributed to floating point / cuDNN nondeterminism rather than any change in the weights themselves.

Per-Class Accuracy

Class	Precision	Recall	F1
water snake	0.76	0.74	0.75
spoonbill	0.90	0.90	0.90
bittern	0.71	0.70	0.71
limpkin	0.83	0.76	0.79
dowitcher	0.88	0.74	0.80
Irish terrier	0.69	0.68	0.69
hyena	0.59	0.82	0.68
monarch butterfly	0.96	1.00	0.98
bighorn sheep	0.78	0.58	0.67
gazelle	0.63	0.52	0.57
acoustic guitar	0.79	0.74	0.76
bathtub	0.78	0.58	0.67
chiffonier	0.92	0.92	0.92
hair slide	0.65	0.56	0.60
marimba	0.53	0.84	0.65
nail	0.48	0.62	0.54
piggy bank	0.58	0.58	0.58
pillow	0.76	0.68	0.72
snowplow	0.79	0.88	0.83
stupa	0.98	0.82	0.89

Confusion Matrix

